CAMPUS FRIEDBERG

MND
Mathematik, Naturwissenschaften und Datenverarbeitung

English translation of the original German document

Examination regulations for the Master's program in Optical System Engineering of the Department 13 Mathematics, Natural Sciences and Data Processing (MND) of the Technische Hochschule Mittelhessen, dated 14 September 2021, Version 3

Approval:

Pursuant to §43 (5) of the Hessian Higher Education Act (HessHG) as amended on December 14, 2021 (GVBl. 2021, p. 931), I hereby approve the following examination regulations for the Master's program in Optical System Engineering.

For the Executive Board:

Giessen, 14 April 2022

Prof. Dr. Matthias Willems,
Presdent of the Technischen Hochschule Mittelhessen

Preliminary remarks:

Pursuant to § 50 (1) No. 1 of the Hessian Higher Education Act (HessHG) in the version of December 14, 2021 (GVBl. 2021, 931), the Department Council of the MND Department of the Technische Hochschule Mittelhessen adopted the examination regulations for the Master's program in Optical System Engineering on September 14, 2021. Part I contains the General Provisions for Master's Examination Regulations of the Technische Hochschule Mittelhessen (THM) dated January 14, 2015 (AMB 01/2015), last amended on April 9, 2025 (AMB 51/2025), and is supplemented by the subject-specific provisions in Part II.

Resolution FBR	Resolution Senate	Approval President	Effective Date / Publication
08 July 2025	03 Sept. 2025	04 Sept. 2025	01 Dec. 2025 / AMB 103/2025
19 Jan. 2023	15 Feb. 2023	19 Feb. 2023	01 June 2023 / AMB 12/2023
14 Sept. 2021	14 Sept. 2021	14 Apr. 2022	01 June 2022 / AMB 24/2022

Part I

General Provisions

The General Provisions for Master's Examination Regulations of the Technische Hochschule Mittelhessen (TUM) published in the Official Gazette of the Technische Hochschule Mittelhessen on January 14, 2015 (AMB 01/2015), last amended on April 9, 2025 (AMB 51/2025), apply.

Part II
Subject-specific provisions

Contents:

§ 1 Scope, Study Objectives

(1) The subject-specific regulations govern the content and requirements of the Master's program in Optical System Engineering of the MND Department and are supplementary to the General Examination Regulations of the Technische Hochschule Mittelhessen.

(2) The objective of this more application-oriented Master's program is to provide students with additional, in-depth, scientifically sound concepts, methods, and techniques after obtaining a first professional university degree, enabling them to apply scientific methods and knowledge in the practical application of optical systems, manufacturing processes, and optical technologies, even to difficult and complex problems.
The Master's program is characterized by:

- scientific orientation,
- holistic treatment of optical systems,
- individual focus,
- emphasis on practical relevance and project orientation,
- development and expansion of teamwork skills, organizational skills, and leadership skills,
- assessment of the social, economic, or cultural impacts of application- or research-related activities and consideration of these in their own actions, as well as
- the Master's thesis as an application-oriented scientific work.

The program prepares students for demanding careers in globally operating companies, in the public sector, or as a self-employed person. It can also serve as the basis for further academic qualification in a subsequent doctoral program.

§ 2 Entry Requirements

(1) The Master's program in Optical System Engineering builds consecutively on a first professionally qualifying university degree with a standard period of study of 7 semesters (210 credit points) from a university in the field of Applied Physics.

The program can also build on a first professionally qualifying university degree with a standard period of study of 7 semesters (210 credit points) in a related degree program. In this case, the first professionally qualifying university degree must contain engineering and physics components, particularly in the field of optics, to approximately the same extent as the Bachelor's program in Applied Physics at the Technical University of Mittelhessen. The Examination Board decides on the fulfillment of this requirement.

(2) Admission to the Master's program in Optical System Engineering is only possible if a first professionally qualifying university degree can be demonstrated in accordance with paragraph 1 and the overall grade of this first professionally qualifying university degree is at least "good" (at least 72.5 percentage points according to § 9 (2) of Part I of the Examination Regulations; 2.5).

If the overall grade is not achieved, the applicant may, upon request, take an examination (aptitude test) to determine whether they have the required aptitude for the Master's program despite not achieving the overall grade.

The aptitude test will be administered in the department. Passing the aptitude test constitutes fulfillment of this admission requirement. Further details are set out in Appendix 6 of the Examination Regulations. Failure to pass the aptitude test means that the required aptitude for the Master's program is not present.

Applicants who have not yet completed their first professionally qualifying university degree by the application deadline may also, upon request, take an examination (aptitude test) in accordance with Appendix 6 if it is sufficiently likely that their final grade will not be at least "good" (2.5; at least 72.5 percentage points according to § 9, (2) of Part I of the Examination Regulations).

(3) If the standard period of study for the first professionally qualifying university degree is less than seven semesters, § 4, (5) must also be observed.

(4) International applicants for the Master's program in Optical System Engineering who have obtained their first professionally qualifying university degree in a non-German-language program must provide evidence of German language proficiency at level A1 of the Common European Framework of Reference for Languages by the end of the enrollment deadline for the Master's program in Optical System Engineering set by the THM at the latest.

§ 3 Admission Requirements

(1) Admission to the Master's program in Optical System Engineering requires a fully completed "Application for Admission to the Master's program in Optical System Engineering" submitted via the portal designated by the THM within the application deadline specified by the THM, including the following documents:

1. Proof of the first professionally qualifying university degree in accordance with § 2, (1 to 3), by:
    - Diploma certificate,
    - Diploma Supplement, and
    - Transcript of Records or comparable proof.

   If these final documents from the first professionally qualifying university degree cannot be submitted by the application deadline, Section 4, Paragraph 7 applies.

2. Description of personal and professional background (CV).

3. Proof of sufficient English language skills at level B2 of the Common European Framework of Reference for Languages, usually demonstrated by the Cambridge First Certificate, the TOEFL test,

the TOEIC test, or the IELTS test. The Admissions Committee decides on the recognition of this proof.

4. For foreign applicants who have obtained their first professional university degree in a non-German-language program: Proof of German language proficiency in accordance with § 2(4). If proof of language proficiency cannot be submitted by the application deadline, § 4, (6) applies.

(2) All documents submitted in accordance with Paragraph 1 must be in German or English. If documents were issued in a language other than German or English, an officially recognized translation of the documents into German or English must be obtained and submitted with the original documents. If documents are issued in English, these documents will be reviewed by the Admissions Committee.

§ 4 Admission Decision

(1) The university decides on the recognition of foreign or equivalent degrees and higher education entrance qualifications in accordance with the statutory provisions and the guidelines of the Standing Conference of the German Rectors and the Ministers of Education and Cultural Affairs of the Länder in the Federal Republic of Germany (Kultusministerkonferenz).

(2) Those who meet the entry requirements according to § 2 and the admission requirements according to § 3 will be admitted to the Master's program in Optical System Engineering. The Examination Board decides on the fulfillment of all subject-specific admission requirements. The Examination Board may involve other professors in the preliminary review of the application documents. In cases of doubt, it may request additional documents from the applicant to make its decision, e.g., certificates of previous relevant professional experience and continuing education.

(3) Admission is granted by the President of the Technische Hochschule Mittelhessen (Admissions Office). Rejection decisions must be communicated to the applicant in writing. The notification must include information on legal remedies.

(4) Applicants who lack the knowledge and skills required for modules in the Master's program in Optical System Engineering from their first professional university degree may be admitted to the Master's program in Optical System Engineering subject to certain conditions. They must acquire the missing knowledge and skills by completing modules from the THM's bachelor's programs, in particular the Bachelor's program in Applied Physics. The admission decision and the determination of the modules to be completed are made by the Examination Board. The missing knowledge and skills may not exceed the 30 credit points that can be acquired. Evidence of the missing knowledge and skills must be provided no later than the time of admission to the Master's thesis.

The modules completed to compensate for this will be certified in the transcript of records. However, they are not part of the Master's program. Earned credit points are not credited towards the Master's program, and the grades achieved are not considered in calculating the final grade.

(5) Bachelor's graduates from programs with a standard period of study of six semesters (180 credit points) may be admitted to the Master's program in Optical System Engineering subject to certain conditions. During the Master's program, they must complete additional modules totaling 30 credit points from the THM's module offerings and demonstrate this by the time they are admitted to the Master's thesis, so that a total study volume of 300 credit points can be demonstrated upon completion of the Master's program in Optical System Engineering. The Examination Board determines the modules to be completed individually based on the course content completed as part of the previous degree.

The additional modules completed are certified in the transcript of records. However, they are not part of the Master's program. Earned credit points are not credited towards the Master's program, and the grades achieved are not considered in the calculation of the final grade.

(6) Applicants who do not meet all of the admission requirements specified in Section 3 by the application deadline may be admitted to the Master's program provided that proof of the missing requirements is provided by the end of the enrollment period for the Master's program in Optical System Engineering set by the THM at the latest. Failure to meet these requirements will result in the cancellation of conditional admission.

(7) Applicants who have not yet completed their first professionally qualifying university degree by the application deadline must submit a detailed certificate of the status of this degree instead of the final documents required in § 3 (1) No. 1 in order to be granted conditional admission to the Master's program.

This certificate must indicate the following:

- all credits achieved so far in the first professionally qualifying degree program,
- the currently achieved and the total credit point score to be achieved; at least 70% of the total CrP to be achieved must already have been earned,
- the current average grade.

By the end of the enrollment period specified by the THM for the Master's program in Optical System Engineering, applicants must provide evidence of completion of their first professionally qualifying university degree with an overall grade of at least "good" (2.5; at least 72.5 percentage points according to Section 9, Paragraph 2 of Part I of the Examination Regulations) or of completion of their first professionally qualifying university degree and passing the aptitude test according to Appendix 6. Failure to comply with these requirements will result in the forfeiture of conditional admission.

§ 5 Enrollment
Admitted applicants must enroll within the deadline set by THM. Enrollment is only possible if all enrollment requirements according to § 3 of the THM Enrollment Regulations, as amended, are met.

§ 6 Master's Degree and Certificate
Upon successful completion of the Master's program in Optical System Engineering, the academic degree "Master of Science," abbreviated to "M.Sc.", is awarded with a certificate as in Appendix 4. The Master's degree opens the door to higher service, to leadership positions in industrial and non-industrial research institutions and companies, in public enterprises and administrations, as well as to career qualification or qualification for higher service or further qualification for doctoral studies.

§ 7 Standard Period of Study, Duration, and Structure of the Program
(1) The standard period of study for full-time study is three semesters, corresponding to 1.5 academic years. To successfully complete the Master's examination, the modules listed in the module overview in Appendix 1 must be successfully completed.

(2) The Master's program can be started in the winter or summer semester.

(3) The duration of the program may be extended by one semester due to additional modules.

§ 8 Modules, Language, Master's Thesis, Colloquium
(1) The modules required must generally be taken from the range of modules offered by the Master's program in Optical System Engineering according to Appendix 1 or 2. Alternatively, identical or equivalent modules from the range of modules offered by other degree programs at the Technische Hochschule Mittelhessen may be taken. Failed attempts will be credited. §§ 11 to 14 of the General

Provisions (Part I of the Examination Regulations) apply.

(2) The Master's program offers six compulsory modules, which students contribute to the curriculum. Four of these modules consist of two sub-modules.

(3) Students can incorporate their personal objectives into the curriculum by selecting their elective modules (see Appendix 1). One of the elective modules taken must be interdisciplinary (soft skill WPF) and thus impart cross-disciplinary qualifications. All other elective modules must be subject-specific.

(4) The language of instruction and examination is generally English. Other languages are specified in the module handbook (see Appendix 2) and will be announced in a timely and appropriate manner at the start of lectures.

(5) The Master's thesis must be written in English or German and include a summary in English. Exceptions will be decided by the Examination Board.

(6) Admission to the Master's thesis is only possible after successfully completing at least 55 credit points from the modules listed in the curriculum (Appendix 1). In the case of a missing module with 6 credit points, the Examination Board will decide on admission to the Master's thesis.

(7) The Master's thesis, including its examination, comprises a total of 30 credit points. The Master's thesis will take 26 weeks, and the thesis will be presented and discussed in a final examination.

(8) In an "accelerated procedure," previously unavailable elective modules that address current topics and are of interest to students may be offered by the department without prior amendment to the examination regulations. The procedural requirements for this are set out in Appendix 2 (Module Handbook). The procedure described above applies accordingly to the placement of elective modules.

(9) For an examination for which the assessment is provided as a written examination, the final repetition may, upon written request from the candidate, be conducted as an oral examination in accordance with Section 7 of the General Provisions. This does not apply to the Master's thesis. The oral examination for a final repetition may only be taken once during the course of study. Each student thus has the opportunity to submit a request for an oral examination for a single module. The Examination Board decides on the request.

§ 9 Fees

No fees are charged for the master's program pursuant to § 20 (5) of the HessHG. The obligation to pay the semester fee pursuant to § 83 (3)of the HessHG, the administrative fee pursuant to § 62 of the HessHG, and fees and contributions pursuant to other statutory provisions remains unaffected.